

Introduction

Prompt Engineering is the practice of designing and refining inputs—known as **prompts**—to guide AI models toward producing useful, accurate, and contextually appropriate outputs. It is based on a simple yet powerful principle: *the quality of AI responses depends directly on the clarity and structure of the prompts provided.*

What is a Prompt?

A **prompt** is the instruction, query, or request you give an AI system. It acts as the foundation of communication between humans and AI.

- **Analogy:** A job briefing or request template—clear instructions lead to high-quality work.
- **Examples:**
 - Poor Prompt: “*Write something about sales.*”
 - Effective Prompt: “*Write a two-paragraph email to a potential client explaining our software’s key benefits for retail businesses.*”

Key Insight: Good prompts reduce back-and-forth effort, save time, and improve accuracy of AI outputs.

Core Components of an Effective Prompt

A well-designed prompt usually includes five essential elements:

1. **Task** – What is to be done (e.g., analyze, summarize, create).
 2. **Instructions** – How it should be done (e.g., bullet points, formal tone).
 3. **Context** – Background information (e.g., audience, purpose).
 4. **Output Template** – Desired structure or format (e.g., Problem–Solution–Next Steps).
 5. **Constraints** – Limitations or requirements (e.g., word count, style, or data sources).
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System Prompts and Role-based Prompting

- **System Prompts** define the AI's *persona* and overall response style, like a director giving background instructions.
 - Example: “*You are a senior business analyst with 10 years of experience. Provide structured, data-driven insights.*”.
 - **Role-based Prompting** assigns the AI a professional perspective (e.g., Project Manager, Customer Service Expert, Financial Analyst), ensuring its responses align with the expectations of that role.
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Prompting Techniques

The lecture introduces a spectrum of techniques tailored to task complexity:

- **Zero-Shot Prompting:** Asking the AI to complete a task without examples. Best for standard business tasks and general knowledge.
 - **Few-Shot Prompting:** Providing 1–3 examples to teach the AI a specific format or style. Useful for consistency in structured outputs.
 - **Chain-of-Thought Prompting:** Adding reasoning cues like “*Let’s think step by step*” to encourage detailed, logical responses.
 - **Prompt Chaining:** Breaking complex tasks into sequential steps, where each output feeds into the next.
 - **JSON Prompting:** Requesting outputs in structured JSON format for easy integration into tools and workflows.
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